Adaptive Repair Scheduling with a Protected Appeal Deadline

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Abstract

An incident desk can attempt optional repairs while a human appeal is pending, provided the repairs leave enough time for the appeal in every compatible duration scenario. For one worker and available nonpreemptive jobs, we give an exact test for starting a repair and a deadline-prefix version for several appeals. For one appeal, a finite policy uses observed job durations to maximize expected benefit under a given prior, or worst-scenario expected benefit without one. Exact checks on synthetic queues compare adaptive policies with fixed schedules and show how separate worst-case durations can rule out a feasible repair. The guarantee requires the realized durations to fit the declared scenario set.

Summary and scope

Question. Which repairs can an incident desk attempt while guaranteeing that protected appeals finish on time?

Finding. A joint duration bound gives an exact reserve test. With several appeals, one earliest-deadline fallback suffices when every deadline prefix fits. For one appeal the finite adaptive recurrence is optimal under a given scenario prior. A separate recurrence handles uncertain priors and shows where randomization can improve the guarantee.

Evidence and scope. The reserve and recurrence are proved. The single-appeal scheduler was checked on 96 synthetic queues, with fixed-sequence and clairvoyant comparisons. The multiple-appeal check concerns the first-move reserve. The guarantee requires one worker, available nonpreemptive jobs, and a duration support that includes the realized scenario.

1 Modeling repairs and protected appeals

A scheduling agent might change several people's appointments through a mistaken interpretation of their preferences. An incident desk then faces a choice among reversals, notifications, and a required appeal. Some reversals lose their value after a time window. The required appeal has a deadline even if several optional repairs could prevent more aggregate harm.

All jobs are available at time zero. One worker executes them nonpreemptively. The job a is a protected appeal, which must finish by d_a in every modeled scenario. For each optional job i, finishing by d_i earns a nonnegative benefit v_i, and finishing later earns zero. Job durations $p_i(z)$ are positive integers in each scenario z from a known finite set with a positive prior. A scenario is fixed for the incident and may correlate all durations. On completion the worker observes that job's duration. Idle time provides no information and cannot increase any benefit in this model. Its omission is without loss because advancing the same sequence weakly improves every deadline.

Stochastic knapsack distinguishes fixed orders from policies that respond to completed work [1]. Here the same unknown scenario governs all job durations. The next repair is chosen only after checking that every compatible scenario leaves a feasible appeal schedule. Benefit then determines the choice among the admissible repairs.

2 The exact reserve test

Let B denote the scenarios compatible with the durations observed so far, and let t be elapsed time. Before the appeal is complete, starting an optional job i is compatible with guaranteed completion of the appeal if and only if

$$t + \max_{z \in B} \{p_i(z) + p_a(z)\} \le d_a.$$

For necessity, any scenario in which this inequality fails would miss the deadline even if the worker immediately switched to the appeal after finishing i. Additional jobs could only delay it. For sufficiency, do the appeal immediately after i in all branches. Starting the appeal itself requires $t + \max_{z \in B} p_a(z) \le d_a$. In a live queue, recompute B from the observed job duration before using the reserve again. A duration outside the modeled support requires escalation, since no scenario then certifies a continuation.

The reserve uses durations that can occur in the same scenario. For example, suppose optional and appeal durations are $(3, 1)$ in one scenario and $(1, 3)$ in the other. The total is four in both. An appeal deadline of four permits doing the optional job first. A reserve computed as $3 + 3$ would prohibit it. When the optional deadline is three and its benefit is four, the joint reserve earns four in either scenario. The marginal reserve must start the appeal and then misses the optional deadline in both scenarios. Pairing the same marginal durations as $(3, 3)$ and $(1, 1)$ instead gives a maximum total of six. The marginal duration laws alone cannot distinguish whether the first repair fits.

3 Several appeals with different deadlines

Let A be the protected jobs still pending. All are available now. An optional job i can be started while preserving a guaranteed fallback exactly when, for every distinct deadline d in A,

$$t + \max_{z \in B} \left(p_i(z) + \sum_{a \in A:\, d_a \le d} p_a(z) \right) \le d.$$

In every scenario the worker must finish all jobs due by d after the optional job, proving necessity. For sufficiency, use the same earliest-deadline order of appeals in every scenario. For each deadline its last job finishes by the corresponding prefix bound. The continuation therefore does not require learning the scenario or choosing a different order for each of them.

A second verifier compared this condition with enumeration of all orders of three protected jobs in 256 synthetic fixtures. Both agreed on every decision. The joint condition admitted 56 first repairs, 25 of which a reserve using the sum of separate duration maxima rejected. Different release times, preemption, or observations during execution would require changing this condition or the scheduling model.

4 Optimal use of completed-job evidence

For a remaining set of jobs R, write $F(R, t, B)$ for the best conditional expected benefit consistent with the hard deadline. For a candidate $i \in R$, each observed duration u defines $B_u = \{z \in B : p_i(z) = u\}$. Let π be the prior, and let $\pi(B)$ denote its mass on the compatible scenarios. Then

$$F(R, t, B) = \max_{i \text{ admissible}} \sum_{u:\, B_u \neq \varnothing} \frac{\pi(B_u)}{\pi(B)} \left(v_i \mathbf{1}\{t + u \leq d_i\} + F(R \setminus \{i\}, t + u, B_u)\right).$$

Set $v_a = 0$. A stop option of value zero is available once the appeal is complete. Before it completes, states without an admissible decision are infeasible.

Theorem 1. *At any feasible initial state, the recurrence returns the maximum expected benefit among adaptive policies that meet the appeal deadline in every modeled scenario.*

Proof. The observed durations restrict the prior to B. Future conditional durations and rewards depend only on R, t, B. The reserve test characterizes which first decisions leave at least one legal continuation. For each possible duration, future decisions can depend on its value, and a legal policy must remain legal in every nonempty branch. Condition on the duration and use induction on $|R|$. Randomizing over admissible first decisions takes an average of their values, so a maximizing deterministic choice is sufficient. □

With n jobs, m scenarios, and maximum duration P, there are at most $2^n(nP + 1)2^m$ states before reachability pruning. This is an exact small-queue method. Its benefit depends on timely duration observations and a useful scenario model. Independent time estimates alone do not identify the joint reserve or the conditional information.

5 Scheduling without a scenario prior

The deadline reserve depends on the duration support. The expected benefit calculation additionally depends on the prior. If probabilities are unavailable, maximize the minimum expected benefit across the listed scenarios. A scenario is fixed before the scheduler draws its coins. The appeal must meet its deadline for every scenario and every policy in the support of that draw.

For a deterministic scheduler, replace the conditional expectation in the recurrence by a minimum over possible observed durations. Within each duration branch, the future guarantee concerns precisely the scenarios compatible with that observation. The policy can choose its continuation separately in every branch. Backward induction therefore gives the best deterministic worst-scenario benefit.

For randomized scheduling, store the attainable reward vectors of deterministic safe policies, with one coordinate per compatible scenario. For each admissible first job, choose one continuation vector for each duration branch. Place it in the coordinates of that branch and add the first job's benefit to each coordinate where it finishes on time. Include the zero vector if the appeal has finished and stopping is allowed.

Theorem 2. *Discarding componentwise dominated vectors at every state preserves the optimal value for an unknown prior. The best randomized worst-scenario value is the largest minimum coordinate of a point in the convex hull of the resulting root vectors.*

Proof. Each safe deterministic policy consists of one admissible first job and one continuation per duration branch. Conversely, any such combination is an implementable policy. Induction

gives the attainable vectors before pruning. Replacing a branch vector with one that dominates it cannot hurt any coordinate, so pruning preserves the best value. Fixing a randomized policy’s coins in advance gives a deterministic policy. Its expected rewards across scenarios therefore form a convex combination of deterministic vectors, and drawing a complete safe policy realizes any such combination. The largest minimum coordinate gives the desired guarantee. □

With two scenarios, it suffices to test the frontier vectors and two-vector mixtures where the two expected rewards are equal. The objective is linear on either side of the diagonal. Each part of the hull has vertices only at original vertices and edge crossings. With more scenarios, the same calculation is a linear program over the retained vectors. The vector sets can grow rapidly, so this remains a small-queue method.

For example, let the appeal take one unit and be due at time one. Two optional repairs each earn ten if finished by time three. Their durations are $(2, 4)$ or $(4, 2)$ in the two possible scenarios. The appeal must run first and reveals nothing about the scenario. Any deterministic next repair earns zero in one scenario. Choose the two repairs with equal probability instead. Each scenario then has expected benefit five, with a timely appeal in every run. The benefit guarantee is over the scheduler’s draw. The appeal guarantee holds on each run.

6 When a small repair is worth doing first

Consider two equally likely scenarios, a protected appeal of duration one due at time three, and a diagnostic repair due at time two. The diagnostic repair takes one or two units and earns one. Two other repairs are due at time five and earn ten each. Their durations are $(2, 5)$ in the first scenario and $(5, 2)$ in the second.

Do the diagnostic repair, do the appeal, and then do the shorter of the two remaining repairs. The diagnostic duration identifies the scenario. Both rewards are earned and the appeal is timely in either case, for an expected benefit of eleven. Any fixed order chooses the same one of the two repairs first. In the scenario where that repair takes five units, finishing by time five requires starting it at zero. That would miss the appeal deadline, so the long repair cannot earn its reward. It also delays the other optional repair beyond its deadline. A fixed order can therefore earn at most ten in only one scenario, plus at most one from the diagnostic repair in each. The order diagnostic, appeal, then the same optional repair in both scenarios attains expected benefit six.

7 When an appeal becomes available later

Suppose each job has a known release time r_i, and its duration in a given scenario stays the same whenever it is started. There are no new observations while the worker idles. Deadlines are unchanged. Under these assumptions, working immediately can be unsafe or can sacrifice a valuable repair. The first-move reserve for jobs that are all available needs a different continuation check.

Theorem 3. *There is an optimal policy that, at each idle decision, starts an available job, stops after completing all protected jobs, or waits until the next release among remaining jobs. The same restriction preserves feasibility and the best known-prior expected benefit with several protected jobs.*

Proof. Take a deterministic policy that meets the deadlines. Given the completed durations, simulate the next job and starting time that this policy would choose. During simulated idle time there is no additional observation, so this choice is known before the next actual action. Execute

that job as early as its release and the current completion time allow. If it is still unavailable, wait through the intervening release events until it is available. Keep simulating the original clock using the same observed job lengths. Inductively the real job starts and finishes no later than in the original policy, in every compatible scenario. All protected deadlines remain satisfied and every earned benefit is preserved. A randomized policy can be treated one fixed coin sequence at a time. With a known prior, an optimal deterministic policy suffices by backward induction over the finite event choices. □

The event-time recurrence uses the earlier state (R, t, B). A start is allowed only after its release and only if every duration branch has a feasible continuation. A protected completion after its deadline makes the branch infeasible. Waiting changes t without changing R or B. There are finitely many decisions because each job completes at most once and time advances to a new release on every wait. The recurrence uses the same conditional expected benefit over completion branches. The left-shift argument also preserves every coordinate in the prior-free reward vectors.

For example, an appeal takes one unit, is released at time one, and is due at time two. A three-unit repair is available at time zero. Starting it misses the appeal. Waiting to do the appeal allows a later one-unit repair, released at time two and due at three, to earn its reward of two. In a second fixture, an immediate repair earns one but occupies the worker until time two. Waiting until time one lets a more valuable repair finish by time two for reward ten, with the appeal still completed by time four.

A separate oracle enumerated timed schedules in two duration worlds, including every integer start time before the last deadline. It accepted a pair only when the schedules made identical choices until a completed duration distinguished the worlds. Its 96 seeded cases agreed with the event recurrence, including 52 cases in which no safe schedule existed. Waiting restored feasibility in seven cases and improved benefit in two further cases compared with a policy that waits only when no job is available. These counts apply to the declared fixture generator and known release dates. Unannounced appeals can provide information while the worker idles, so they require a different observation model.

8 Finite verification and operational use

The verifier generated 96 instances with five optional jobs, one protected appeal, and four equally likely duration scenarios. It replayed every selected adaptive policy in all 384 scenarios and checked both the protected deadline and the reported benefit. It also enumerated fixed sequences and separately computed a clairvoyant upper bound, where the scheduler is told the scenario before choosing its order.

Scheduler	Sum of expected benefits over 96 instances
Best fixed sequence	1,884.25
Adaptive with marginal reserve	2,033.00
Adaptive with joint reserve	2,039.75
Clairvoyant upper bound	2,151.50

The joint adaptive scheduler improved on the fixed order in 80 instances and on its marginal-reserve counterpart in three. The largest expected improvement over a fixed order was 5.75 units. The synthetic weights are decision inputs, not measurements of real harm.

The prior-free verifier independently enumerated legal job orders in each scenario and paired only orders that agree until their observed durations diverge. The resulting nondominated reward

vectors agreed with the recurrence on 96 two-scenario queues. Randomization strictly improved the deterministic worst-scenario benefit in one of these fixtures, in addition to the constructed two-repair witness. These counts describe the fixed test generator.

A service could use the recurrence in a short repair queue when a pause or reversal can reveal whether a dependency is still blocked. The required response time and the category of protected case should be determined by the incident process. If an observed duration rules out every scenario, the planned reserve no longer applies. A separate human escalation route must handle that case. A prospective drill should hold the worker budget fixed, replay correlated outages, and compare prevented harm, protected deadline misses, and sensitivity to unmodeled durations.

An initial hand-calculated expectation for the reserve witness was corrected before the successful check. The change was confined to the expected assertion.

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References

[1] Brian C. Dean, Michel X. Goemans, and Jan Vondrák. *Approximating the Stochastic Knapsack Problem: The Benefit of Adaptivity.* Mathematics of Operations Research 33(4), 2008. `https://math.mit.edu/~goemans/PAPERS/DeanGoemansVondrak-2008-ApproximatingTheStochasticKnapsackProblem.pdf`.